July 30, 2026

Dr. Christopher Hirata Department of Physics / Center for Cosmology and AstroParticle Physics The Ohio State University [street address]

Dear Dr. Hirata,

I'm an independent researcher, and my first job in this letter is to temper the skepticism an outsider's package rightly earns. That skepticism is warranted, and I've faced it from the start. I learned to avoid the crank-trap the direct way, by falling into it early. I call that my post-graduate phase, and I'll confess it freely. What came out of it is a working protocol, and it reduces to three rules.

Transparency. From first philosophy to final prediction, all of it is open source. Every commit is tracked, every derivation repeatable, every route tried, every mistake, retraction, and staleness flag recorded alongside the results. The failures sit in the repo next to the wins.

Honest-tiering. An outsider has to be as loud about what is *not* claimed as about what is. Every line in the tables that follow carries one of 19 rankings (Derived, Measured, Postulated, Inherited, Asserted, and so on), each verifiable against the paper it cites. Nothing is presented as proven that isn't, and the walls and open problems are labeled as exactly that.

Adversarial review. The physics is mine. AI has been a productive tool, and like any tool it is fallible and prone to its own kind of drift, which is a fair source of skepticism in itself. So I use it deliberately as a breaker as well as a builder: every paper in the repository is handed to a model calibrated to dismantle it, and I work across several models that check one another rather than trusting any single one. The method is written up in the repo for anyone who wants it.

With that baseline established, and hopefully a little of your serious attention earned, here is the claim and the case for what I call Event Density: a unified framework for physics built from one small set of substrate primitives.

It carries a parameter-free cosmological prediction that should interest you directly: the MOND acceleration scale is not a constant of nature but tracks the Hubble rate, $\mathrm{a}_0(\mathrm{z}) = \mathrm{cH}(\mathrm{z})/2\pi$, a clean redshift-dependent signature for exactly the rotation-curve and weak-lensing surveys you work with. The first data, MUSE-DARK III in 2026, already shows the scale evolving at about 30σ. Closer to your lensing work, the same structure forbids gravitational slip beyond $\mathrm{O}(\Phi)$, with no vector field to repair it, a specific null a metric-only coupling cannot dodge.

The enclosed pages open with three short reference tables, in a deliberate order.

The **Core Predictions come first**, ahead of the theory. An outside idea has to make itself vulnerable before it asks to be believed. Falsification and novelty are the skeptic's final recourse, so I lead with exactly where Event Density diverges from MOND, from ΛCDM, and from standard quantum theory, and with the measurements that would kill it.

The **Core Theory** follows as a single table: 13 primitives (the axioms), 10 constants (4 of them derived from the others, marked $*$), and 15 postulates. Thirty-eight lines in total. That is the entire input.

The **Core Claims** come last of the three: the 159 derivations and groundings those 38 lines reach across a dozen domains of physics. The full ledger, including every assertion, inheritance, and open problem, lives in the repository.

A longer **Report** in the repo gives the narrative behind these tables, and it in turn points to the individual papers where each claim is worked out in full. And last, though it sits most prominent in my own mind, I've enclosed a short **essay** on the consilience itself: how few the primitives are, and how far they reach.

If this holds up to your reading and you think it merits another set of eyes, please feel free to pass it to David Weinberg or Annika Peter. The dark-energy and dark-matter predictions bear on their work as much as the cosmology bears on yours.

I'm not asking you to believe any of it. I'm asking you to check it where it says it can be broken. The repository, with timestamped DOIs for every result, is at [URL], and the QR code below goes to the same place.

With respect, and thanks for your time,

Allen Proxmire [contact] · [repo URL] · [QR]